DE LA SALLE UNIVERSITY-DASMARINAS
COLLEGE OF INFORMATION AND COMPUTER STUDIES
INFORMATION TECHNOLOGY DEPARTMENT

COURSE SYLLABUS

First Semester, SY 2024-2025

COURSE CODE	:	S-ITPE004LA
COURSE TITLE	:	IT PROFESSIONAL ELECTIVE 4 (SYSTEM INTEGRATION AND ARCHITECTURE 2 LEC)
COURSE TYPE	:	Lecture
COURSE CREDIT	:	Two Units
LEARNING MODALITY	:	Traditional Blended Learning
PRE-REQUISITES	:	SYSTEM INTEGRATION AND ARCHITECTURE 1
CO-REQUISITES	:	SYSTEM INTEGRATION AND ARCHITECTURE 2 LAB
PROFESSOR	:	MARYLI F. ROSAS, DIT
CONSULTATION HOURS	:	TUES 7:00-8:00 & 10:00-12:00; WED 7:00-12:00; FRI 2:00-4:00 COS100C
CONTACT INFORMATON	:	mfrosas@dlsud.edu.ph (046) 4811900 local 3134

VISION

A leading Catholic university that inspires excellence and drives innovation toward a just, peaceful, and sustainable society.

MISSION

To champion the Human and Christian Education of lifelong learners who value history and culture through responsive and inclusive academic, research, and extension programs.

COURSE DESCRIPTION:

This course focuses on project management, testing and quality assurance, and organizational context architecture. The course introduces the students on usability testing and recommends appropriate actions based on the results of the test. Furthermore, topics on the construction of an architectural model of a complex system using an architectural framework are discussed.

This course builds upon the fundamentals of system architecture and integration, exploring advanced concepts and techniques for designing, integrating, and managing complex systems. Emphasis is placed on real-world applications, case studies, and hands-on projects.

COURSE LEARNING OUTCOMES:

By the end of this course, students can:

CLO 1. evaluate software requirements and appraise the processes involved in discovering and documenting these requirements;
CLO 2. examine different system models that may be developed as part of requirements engineering and system design processes;
CLO 3. analyze the concepts of software architecture and architectural design;
CLO 4. evaluate object-oriented software design using the UML and highlight important implementation concerns;
CLO 5. identify software testing and software testing processes;
CLO 6. explain why software evolution is such an important part of software engineering and to describe the challenges of maintaining a large base of software systems, developed over many years;
CLO 7. discuss software project management and two important management activities, namely, risk management and people management; and
CLO 8. demonstrate 21st century skills in all learning activities.

LEARNING PLAN:

Learning Outcomes for Midterm Period

Course Learning Outcomes	Topic Learning Outcomes
CLO 1. Evaluate software requirements and appraise the processes involved in discovering and documenting these requirements.	• TLO 1. Evaluate the concepts of user and system requirements and why these requirements should be written in different ways. • TLO 2. Value the differences between functional and nonfunctional software requirements. • TLO 3. Appraise the main requirements engineering activities of elicitation, analysis, and validation, and the relationships between these activities. • TLO 4. Weigh why requirements management is necessary and how it supports other requirements engineering activities.
CLO 2. Examine different system models that may be developed as part of requirements engineering and system design processes.	• TLO 5. Relate how graphical models can be used to represent software systems and why several types of models are needed to fully represent a system. • TLO 6. Analyze the fundamental system modeling perspectives of context, interaction, structure, and behavior. • TLO 7. Distinguish the principal diagram types in the Unified Modeling Language (UML) and how these diagrams may be used in system modeling. • TLO 8. Relate to model-driven engineering, where an executable system is automatically generated from structural and behavioral models.
CLO 3. Analyze the concepts of software architecture and architectural design.	• TLO 9. Distinguish why the architectural design of software is important. • TLO 10. Examine the decisions that must be made about the software architecture during the architectural design process. • TLO 11. Relate to the idea of Architectural patterns, well-tried ways of organizing software architectures that can be reused in system designs. • TLO 12. Distinguish how Application-Specific Architectural patterns may be used in transaction processing and language processing systems.

CLO 4. Evaluate object-oriented software design using the UML and highlight important implementation concerns.	• TLO 13. Appraise the most important activities in a general, object-oriented design process. • TLO 14. Select some of the different models that may be used to document an object-oriented design. • TLO 15. Support the idea of design patterns and how these are a way of reusing design knowledge and experience. • TLO 16. Weigh key issues that must be considered when implementing software, including software reuse and open-source development.
CLO 8. Demonstrate 21st century skills in all learning activities.	• *Collaboration, Skilled Communication, Knowledge Construction, Self-Regulation, Real-World ProblemSolving, Use of ICT (not part of topic, but integrated in assessments)*

Module No and Learning Outcomes	**Week**	**Teaching-Learning Activities / Assessment Strategies**	**Technology Enabler**	**Onsite / F2F**	**Asynchronous**	**Allotted Hours**
Module 1 TLO 1 TLO 2 TLO 3 TLO 4 CLO 1 CLO 8	Week 1 Aug 19-24	• Gospel Reading, Reflection, and Prayer • Syllabus uploaded. • Presentation of course introduction • Discussion of Module 1 – Requirements Engineering • EA: Assignment 1: Submit the Use Case of your Thesis	MS Teams PowerPoint Schoolbook	✓		2
	Week 2 Aug 26-31	• Gospel Reading, Reflection, and Prayer • Continuation of Discussion of Module 1 – Requirements Engineering • EA: Short Quiz on Chapter 1	MS Teams PowerPoint Schoolbook		✓	2
Module 2 TLO 5 TLO 6 TLO 7 TLO 8 CLO 2 CLO 8	Week 3 Sept 2-7	• Gospel Reading, Reflection, and Prayer • Discussion of Module 2 – System Modeling Context models Interaction models Structural models Behavioral models Model-driven engineering	MS Teams PowerPoint Schoolbook	✓		2

	Week 4 Sept 9-14	• Gospel Reading, Reflection, and Prayer • Online Examination Covering Module 1 and Module 2	MS Teams PowerPoint Schoolbook		✓	2
	Week 5 Sept 16-21	• Gospel Reading, Reflection, and Prayer • Discussion of Module 3 – Architectural Design • Seatwork: Draw diagrams showing a conceptual view and a process view of the architectures of your thesis.	MS Teams PowerPoint Schoolbook	✓		2
Module 3 Module 4 TLO 9 TLO 10 TLO 11 TLO 12 TLO 13 TLO 14 TLO15 TLO 16 CLO 3 CLO 4 CLO 8	Week 6 Sept 23-28	• Gospel Reading, Reflection, and Prayer • Continuation of Discussion of Module 3 – Architectural Design • EA: Online Quiz on Module 3 • Journal Article: Papakostas, C., Troussas, C., Sgouropoulou, C. (2024). Artificial Intelligence-Enhanced PARSAT AR Software: Architecture and Implementation. In: Special Topics in Artificial Intelligence and Augmented Reality. Cognitive Technologies. Springer, Cham. https://doi.org/10.1007/978-3-031-52005-1_5	MS Teams PowerPoint Schoolbook		✓	2
	Week 7 Sept 30 - Oct 5	• Gospel Reading, Reflection, and Prayer • Discussion of Module 4 – Design and Implementation • EA: Online Quiz on Module 4	MS Teams PowerPoint Schoolbook	✓		2
	Week 8 Oct 7-12	• Gospel Reading, Reflection, and Prayer • Checking of Thesis Progress Report (Software and Documentation)	PowerPoint Schoolbook		✓	2
	Week 9 Oct 14-19	MIDTERM EXAMINATION		✓		2

TOTAL	10	8	18

Learning Outcomes for Final Period

Course Learning Outcomes	Topic Learning Outcomes
CLO 5. Comprehend software testing and software testing processes.	• TLO 17. Understand the stages of testing from testing during development to acceptance testing by system customers. • TLO 18. Identify techniques that help us choose test cases that are geared to discovering program defects. • TLO 19. Recognize test-first development, where you design tests before writing code and run these tests automatically. • TLO 20. Discuss the three distinct types of testing—component testing, system testing, and release testing • TLO 21. Explain the distinctions between development testing and user testing.
CLO 6. Explain why software evolution is such an important part of software engineering and to describe the challenges of maintaining a large base of software systems, developed over many years.	• TLO 22. Recognize that software systems must adapt and evolve if they are to remain useful and that software change and evolution should be considered as an integral part of software engineering. • TLO 23. Discuss what is meant by legacy systems and why these systems are important to businesses. • TLO 24. Describe how legacy systems can be assessed to decide whether they should be scrapped, maintained, reengineered, or replaced. • TLO 25. Recognize different types of software maintenance and the factors that affect the costs of making changes to legacy software systems.
CLO 7. Discuss software project management and two important management activities, namely, risk management and people management.	• TLO 26. Define the principal tasks of software project managers. • TLO 27. Explain risk management and some of the risks that can arise in software projects. • TLO 28. Discuss factors that influence personal motivation and what these might mean for software project managers. • TLO 29. Recognize key issues that influence team working, such as team composition, organization, and communication.
CLO 8. Demonstrate 21st century skills in all learning activities.	• *Collaboration, Skilled Communication, Knowledge Construction, Self-Regulation, Real-World Problem-Solving, Use of ICT (not part of topic, but integrated in assessments)*

Module No and Learning Outcomes	**Week**	**Teaching-Learning Activities / Assessment Strategies**	**Technology Enabler**	**Onsite / F2F**	**Asynchronous**	**Allotted Hours**
Module 5 Module 6 TLO 17 TLO 18	Week 10 Oct 21-26	• Gospel Reading, Reflection, and Prayer • Discussion of Module 5 – Software Testing	MS Teams PowerPoint Schoolbook	✓		2

TLO 19 TLO 20 TLO 21 TLO 22 TLO 23 TLO 24 TLO 25 CLO 5 CLO 6 CLO 8		• Journal Article: Rosas, M. F., Ambat, S. C., & Lagman, A. C. (2019). Extraction of students' feedback in university services using logistic regression and chi-square automatic interaction detection (chaid) algorithm. *International Journal of Simulation--Systems, Science & Technology*, *20*(2). • EA: Individual Seatwork: Based on your experience with a bank ATM, draw an activity diagram that models the data processing involved when a customer withdraws cash from the machine.				
	Week 11 Oct 28 -Nov 2	***Independent Learning Week***			✓	2
	Week 12 Nov 4-9	• Gospel Reading, Reflection, and Prayer • Continuation of Module 5 – Software Testing Journal Article : M. F. Rosas, S. C. Ambat, A. D. Magnaye and J. R. F. Rosas, "Classification and regression decision tree: a mining technique for students' insights on the university services with text analysis," 2018 9th IEEE Annual Ubiquitous Computing, Electronics & Mobile Communication Conference (UEMCON), New York, NY, USA, 2018, pp. 971-976, doi: 10.1109/UEMCON.2018.8796828. M. F. Rosas, S. C. Ambat and M. A. Ballera, "Data mining of students' response on the university services using chi-square automatic interaction detector (CHAID) algorithm," 2018 1st IEEE International Conference on Knowledge Innovation and Invention (ICKII), Jeju, Korea (South), 2018, pp. 244-247, doi: 10.1109/ICKII.2018.8569209.	MS Teams PowerPoint Schoolbook	✓		2
	Week 13 Nov 11-16	• Gospel Reading, Reflection, and Prayer • EA: Online Quiz on Module 5	MS Teams PowerPoint Schoolbook		✓	2

	Week 14 Nov18-23	• Gospel Reading, Reflection, and Prayer • Self-paced reading: Module 6 – Software Evolution	Schoolbook		✓	2
Module 6 Module 7 TLO 22 TLO 23 TLO 24 TLO 25 TLO 26 TLO 27 TLO 28 TLO 29 CLO 6 CLO 7 CLO 8	Week 15 Nov 25-30	• Gospel Reading, Reflection, and Prayer • Continuation of Module 6 – Software Evolution • EA: Assignment: Exercises (by group) • Journal Article: Khojah R, Mohamad M, Leitner P, and Neto FG. 2024. Beyond code generation: an observational study of chatgpt usage in software engineering practice. Proc. ACM Softw. Eng. 1, FSE, Article 81 (July 2024), 22 pages. https://doi.org/10.1145/3660788	PowerPoint Schoolbook	✓		2
	Week 16 Dec 2-7	• Gospel Reading, Reflection, and Prayer • EA: Examination Covering Module 5 and 6	MS Teams		✓	2
	Week 17 Dec 9-14	• Gospel Reading, Reflection, and Prayer • Discussion of Module 7 – Project Management • EA: Short Quiz on Chapter 7 • Journal Article : Daraojimba EC., Nwasike CN, Adegbite AO, Ezeigweneme CA, & Gidiagba JO. (2024). Comprehensive review of agile methodologies in project management. Computer Science & IT Research Journal, 5(1), 190-218. https://doi.org/10.51594/csitrj.v5i1.717	PowerPoint Schoolbook	✓		2
	Week 18 Dec 16-18	• Gospel Reading, Reflection, and Prayer • FINAL EXAMINATION: Onsite Examination Covering Module 7		✓		2
TOTAL				10	8	18

GRADING SYSTEM:

Midterm	%	Final	%
Enabling Assessments	50%	Enabling Assessments	50%
Summative Assessment	30%	Summative Assessment	30%
Class Participation	20%	Class Participation	20%
	100%		100%

FINAL GRADE = $\frac{\text{Midterm Grade + Final Grade}}{2}$

COURSE POLICIES AND REQUIREMENTS:

1. This course follows the Traditional Blended Learning Mode. There will be 50% onsite and 50% asynchronous.

2. Enrollment in an E-Class. The Center for Innovative Learning Programs (CILP) shall enroll you in the e-class. This will be based on the class list provided by the Office of the University Registrar and at the portal.

3. Communication and Feedback
    a. Onsite classes shall happen as indicated in the learning plan section of the syllabus. Fifty percent (50%) of the weekly classes is delivered face-to-face while the remaining is asynchronous.
    b. Online Asynchronous Communication shall happen when the faculty utilizes at least one of the recommended asynchronous communications. The minimum response time for asynchronous communication is 48 hours, excluding Sundays and holidays. All communication platforms must be explicitly indicated in the syllabus. Below are the recommended asynchronous communication platforms:
        i. Forum – at the Forum Tab at Schoolbook, students can post their general concerns. The forum is public, and any post can be seen by the class.
        ii. Messaging – at Schoolbook. Any private or confidential matters may be messaged through the Schoolbook messaging feature.
        iii. Email – you may email your professor. (Professor's email will be given during the course orientation)
    c. No use of foul, curse, and other bad words, whether written or verbal. No shouting. No flooding of chat and messages. No selling.

4. Attendance Policy
    a. Attendance Policy for Onsite Classes is scheduled based on the learning plan. A maximum of 20% for class participation/recitation/attendance shall be allotted for the student's grade.
    b. The onsite meeting will only be held at the schedule indicated in the registration form.
    c. Students are required to attend onsite classes, 40% of absence (from onsite classes) for the entire semester means 0.0 final grade.

5. Intellectual Property Policy
    a. Policy and copyright infringement will not be tolerated. Any violation of such will earn students a grade of zero in the assessment.
    b. Using any illegally obtained software is strictly prohibited.
    c. Students are encouraged to use their original photos, videos, and other resources. Otherwise, students can use royalty-free resources or embed the sources in their submissions to avoid copyright infringement and/or plagiarism.
    d. Giving of passwords to Schoolbook and Office 365 is strictly prohibited. Likewise, accessing Schoolbook and Office 365 accounts other than the students' personal account is also strictly prohibited. Violating students will be reported to the Student Welfare and Formation Office (SWAFO).

6. Other Policies
    a. Attire - students must wear decent casual or smart-casual attire prescribed by the Student Handbook.
    b. Students must be honest at all times; cheating and plagiarism in any form will merit a grade of 0.00.
    c. Cellular/Mobile phones should always be in silent mode during class hours; the use of cellular phones is prohibited in class unless special permission is sought. Cellular phones cannot also be used as calculators during examination.
    d. Any complaints against the teacher or against classmates should be properly addressed to the subject teacher for appropriate action. Students may seek the help and guidance of their academic/registration adviser in resolving the issue with the subject teacher when such mediation is needed.

REFERENCES:

On-Site References

Provider	Reference Material
AEA Lib	Leach, Ronald J. (2016). Introduction to Software Engineering. 2nd Edition. Boca Raton. CRC Press, Taylor & Francis Group
AEA Lib	Pressman, Roger S. (2015). Software engineering: a practitioner's approach. Singapore: McGraw-Hill Book Co.

Online References

Call Number/ E-provider	Reference Material
e-Book (downloaded)	Sommerville, Ian. (2017). Software Engineering. 10th Global Edition. Pearson Education Ltd.
Web Resource	Alliance Software. (2020). An Introduction to Software Development Methodologies. [Accessed : 2024 August 16]. https://www.alliancesoftware.com.au/introduction-software-development-methodologies/
Web Resource	Muhammad Raza. (2020). The Software Development Life Cycle: An Introduction. [Accessed : 2024 August 16]. Retrieved from https://www.bmc.com/blogs/sdlc-software-development-lifecycle/
Web Resource	Carriou Erwan. (2019). Designing System. [Accessed : 2024 August 16] https://medium.com/swlh/designingsystemsce28ee31c0cd
Web Resource	Study.com (2021). Video Team: Requirements Modelling in Software Engineering. [Accessed : 2024 August 16] https://study.com/academy/lesson/requirements-modeling-in-software-engineering-classesfunctionsbehaviors.html
Web Resource	Guru99. (2020) What is System Testing? Types and Definition. [Accessed : 2024 August 16] https://www.guru99.com/system-testing.html

Prepared: **Information Technology Department**
1st Semester 2024-2025

Approved: **(SGD) Sherry Naz**
Chair, ITD

Endorsed: **(SGD) Marivic Mitschek**
Dean, CICS